7. What is a photo cell? Mention the different types of photocells.
8. Write the expression for the de Broglie wavelength associated with a charged particle of charge q and mass m, when it is accelerated through a potential V.
9. State de Broglie hypothesis.
10. Why we do not see the wave properties of a baseball?
11. A proton and an electron have same kinetic energy. Which one has greater de Broglie wavelength? Justify.
12. Write the relationship of de Broglie wavelength λ associated with a particle of mass m in terms of its kinetic energy K.
13. An electron and an alpha particle have same kinetic energy. How are the de Broglie wavelengths associated with them related?
14. Define stopping potential.
15. What is surface barrier?
16. Mention the two features of x-ray spectra, not explained by classical electromagnetic theory.
17. What is Bremsstralung?

III Long Answer Questions

1. What do you mean by electron emission? Explain briefly various methods of electron emission.
2. Briefly discuss the observations of Hertz, Hallwachs and Lenard.
3. Explain the effect of potential difference on photoelectric current.
4. Explain how frequency of incident light varies with stopping potential.
5. List out the laws of photoelectric effect.
6. Explain why photoelectric effect cannot be explained on the basis of wave nature of light.
7. Give the quantum concept of energy proposed by Max Planck.
8. Obtain Einstein's photoelectric equation with necessary explanation.
9. Explain experimentally observed facts of photoelectric effect with the help of Einstein's explanation.
10. Give the construction and working of photo emissive cell.
11. Derive an expression for de Broglie wavelength of electrons.
12. Briefly explain the principle and working of electron microscope.
13. Describe briefly Davisson – Germer experiment which demonstrated the wave nature of electrons.
14. List out the characteristics of photons.
15. Give the applications photocell.
16. How do we obtain characteristic x-ray spectra?

IV. Numerical problems

1. How many photons per second emanate from a 50 mW laser of 640 nm?

[Ans: $1.61\times10^{17}\,s^{-1}$]

2. Calculate the maximum kinetic energy and maximum velocity of the photoelectrons emitted when the stopping potential is $81V$ for the photoelectric emission experiment.

[Ans: 1.3×10^{-17}J; $5.3\times10^{6}\,ms^{-1}$]

3. Calculate the energies of the photons associated with the following radiation: (i) violet light of 413 nm (ii) X-rays of 0.1 nm (iii) radio waves of 10 m.

[Ans: 3 eV; 12424 eV; 1.24×10^{-7} eV]

XII_Physics_Vol-2_UNIT-8_EM.indd 137 03-03-2022 15:21:10